Why zero is an informational singularity

Information Theory | Version 1.8

ZP Companion | April 2026

This companion explains the ideas in plain language with diagrams and real-world examples. It is not the formal document — every claim here restates a result already proved in the corresponding technical document. Consult that document for the authoritative mathematics.

What Is ZP-C Doing?

ZP-C establishes that zero is informationally unreachable — using two completely independent measures that arrive at the same conclusion from different starting points.

Route 1 — Kolmogorov complexity (algorithmic): How long must a program be to describe a configuration? At the incompressibility threshold P_0, no shorter description exists — the string must be specified in full, every bit. Algorithmic depth is unbounded.

Route 2 — 2-adic surprisal (probabilistic): How much probability mass does a state carry? As a state approaches zero in $\mathbb{Q}_2$, its probability approaches zero, and its surprisal $I(x) = -\log_2 P(x)$ approaches infinity. Informational cost is unbounded.

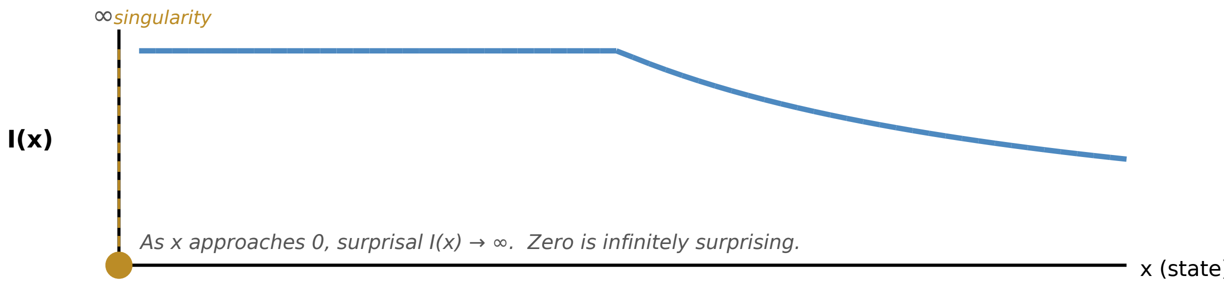
Neither measure knows about the other. K counts program lengths; I(x) counts probability. Both go to infinity at the same threshold. ZP-C operates in $\mathbb{Q}_2$ (from ZP-B) throughout — smooth calculus tools are not valid on a totally disconnected space.

Real-world example — ZIP compression hitting a wall

Some files compress a lot; others barely at all. A file that cannot be compressed further has hit its incompressibility threshold — Route 1. A truly random file also has maximum surprisal: every bit is equally likely, nothing is predictable — Route 2. Both descriptions identify the same extreme: the point where no shortcut exists.

Surprisal: The Cost of a Transition

The surprisal of state x is $I(x) = -\log_2 P(x)$: how surprising it is to observe x. As x approaches 0, P(x) approaches 0, and I(x) approaches infinity. Zero is infinitely surprising — it cannot be reached by any path of finite informational cost.



Surprisal I(x) as x approaches 0 (amber singularity). Every path toward 0 accumulates unbounded informational cost.

Real-world example — A perfectly random password

To describe a truly random password, you have to send every character — you can't summarize it. That's maximum surprisal: the information content equals the full length of the string. Zero in $\mathbb{Q}_2$ is the limit of this — infinite depth, infinite surprisal.

Where the Two Routes Converge

Kolmogorov complexity K and 2-adic surprisal I(x) are genuinely different tools. K measures how long a program must be to reproduce a string — it is a property of algorithms. I(x) measures how rare a state is — it is a property of probability distributions. They share no common definition and were developed in entirely separate branches of mathematics.

Yet both diverge to infinity at the same point: zero. $K(x|n)/n \to 1$ as configurations approach the incompressibility threshold. $I(x) \to \infty$ as states approach 0 in $\mathbb{Q}_2$. The convergence is not engineered — it reflects something structurally real about ⊥. Zero is not merely hard to reach in one sense; it is the limit of unreachability from two independent directions simultaneously. ZP-C uses both routes as independent confirmation of the same structural fact.

Why the Singularity Forces Execution (L-INF)

The surprisal graph shows that I(x) goes to infinity as x approaches 0. ZP-C v1.5 makes this precise with Lemma L-INF (Informational Extremity of ⊥): at the incompressibility threshold P_0, the configuration's surprisal is not just very large — it is formally infinite. The branching measure assigns probability approaching zero to any specific configuration at P_0, so $I(P_0) = \infty$.

This matters because infinite surprisal means no finite external program can bound the informational content of the configuration. A static stored description always has finite content. So a configuration at P_0 cannot be a stored description — it must be something actively running. L-INF is the formal reason P_0 forces execution, and it is what connects the surprisal singularity to the L-RUN argument below.

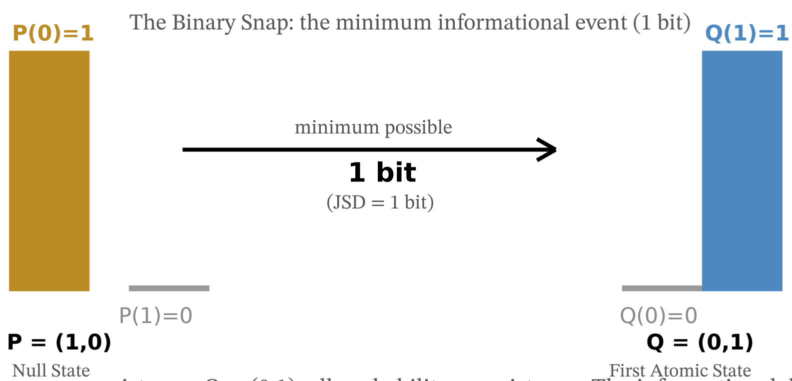
The branching measure — the way ZP-C assigns probabilities to states in D4 — is a representational commitment (RP-2 in ZP-C v1.6). It is well-motivated by the binary structure of AX-B1, but it is a choice, not a derivation. The framework labels it explicitly so readers know where a design decision is being made.

Analogy — A file that cannot be described

A truly incompressible file has no pattern — every bit is essential, nothing can be summarized or shortened. At P_0, the configuration is like this file: no shorter description exists. The only way such a thing can "be present" is as something actively running — not a stored record waiting to be read.

The Binary Snap Costs Exactly 1 Bit

The Jensen-Shannon Divergence (JSD) between the Null State P = (1,0) and the First Atomic State Q = (0,1) is exactly 1 bit. These distributions are derived from AX-B1 — not assumed. The Snap is the minimum informational event possible.



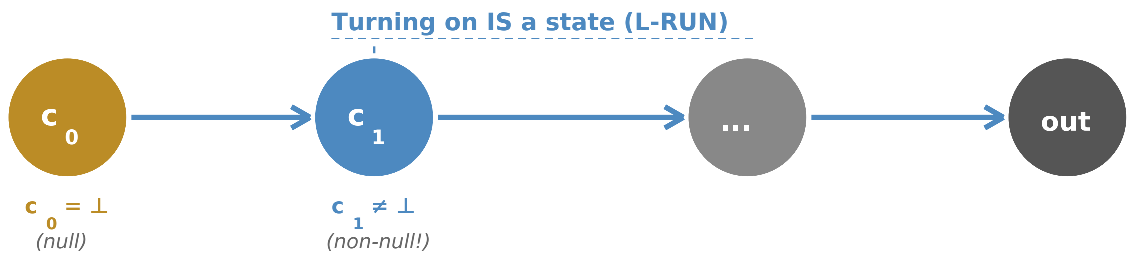
P = (1,0): all probability on non-existence. Q = (0,1): all probability on existence. The informational distance is exactly 1 bit — the minimum possible transition.

Turning On Is a State (L-RUN)

Can a program output the null state ⊥ without passing through any non-null intermediate state? The answer is no — because the act of turning on is itself a state.

Any program that executes must pass through a "first instruction fetched" configuration (c_1). That configuration is distinct from the not-yet-running state (c_0). By AX-B1 it is non-null. The output being null does not mean the intermediate states were null.

The identification of c_0 with the null state ⊥ is a modeling commitment (CC-2 in ZP-C v1.6) — a choice made explicit in the formal document, parallel to CC-1 in ZP-A (which identifies the initial state S_0 with ⊥). It is not derived from the machine definition; it is chosen. Labeling it CC-2 keeps that choice visible.



L-RUN: the transition $c_0 \to c_1$ is a non-null state change. Execution itself is a state — independent of output.

Real-world example — Booting a computer

When you press the power button, the machine doesn't jump from "off" to "showing your desktop." It passes through BIOS, POST, bootloader, kernel — each a distinct non-null configuration state. Even if programmed to immediately wipe itself and show nothing, it still passed through those intermediate states.

Remember: Computers illustrate the result. L-RUN applies to any Turing machine — the abstract mathematical model of computation. The conclusion is mathematical, not technological.

New in v1.6: Two Named Commitments

ZP-C v1.6 adds explicit labels to two choices the framework makes. Labeling them is not a weakness — it is how the framework keeps track of what is proven versus what is chosen.

CC-2 (Modeling Commitment): The Turing machine initial configuration c_0 is identified with the null state ⊥. This is the same pattern as CC-1 in ZP-A, which identifies the initial state S_0 with ⊥. Neither identification is forced by the definitions — both are deliberate choices that make the multi-layer framework cohere.

RP-2 (Representational Commitment): The branching measure used to assign probabilities to states is a representational choice. It is the natural choice given a binary state space, but it is not the only valid option. Labeling it RP-2 makes the commitment visible to anyone evaluating the framework's assumptions.

Key Result: T-SNAP — The Binary Snap Is a Proven Theorem

The derivation chain: the incompressibility threshold P_0 produces a configuration with infinite surprisal (L-INF). A configuration with infinite surprisal cannot be a static stored description — it must be a live computation (DA-1, ZP-E). Any live computation must execute a first instruction, and that execution is a non-null state (L-RUN). No Turing machine program can produce any output without passing through such a non-null intermediate state. And any non-null state change starting from ⊥ is precisely the Binary Snap (ZP-A D2). ZP-E closes the chain as Theorem T-SNAP. The Binary Snap is no longer assumed — it is derived.